

# Report R&D

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**Problem Statement:** Complex Microstructural identification in metallic alloys using computer vision, particularly Deep Learning (CNNs)

**Research paper referred:**

[https://www.sciencedirect.com/science/article/pii/S0927025620300847?ref=pdf\\_download&fr=RR-2&rr=9cc65ee2fdbf48d4](https://www.sciencedirect.com/science/article/pii/S0927025620300847?ref=pdf_download&fr=RR-2&rr=9cc65ee2fdbf48d4)

## Abstract

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This project focuses on the automated characterization of material microstructures using a two-stage machine learning pipeline. The system combines image classification using Convolutional Neural Networks (CNN) and feature extraction through image segmentation techniques. The dataset consists of micrographs of Ti-6Al-4V alloy, classified into three categories: lamellar, duplex, and acicular. The trained CNN achieves an accuracy of approximately 94%, and segmentation techniques such as watershed algorithm and Histogram of Oriented Gradients (HOG) are used to extract meaningful quantitative features. This approach reduces manual effort and improves efficiency in material characterization. The segmented image is passed into other code and the fraction of lamellar grains/duplex grains in that particular image.

## Problem Statement Analysis

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Material science increasingly relies on data-driven approaches to understand relationships between processing, structure, and properties.

**Manual microstructure analysis:**

- Time-consuming
- Subjective
- Not scalable

**Objective:**

Develop a system that:

1. Classifies microstructure images
2. Extracts meaningful features automatically

## Dataset Description

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The images were acquired from SEM (scanning electron microscope)

- **Material:** Ti-6Al-4V alloy
- **Total images:** 1225 micrographs

- **Image size:** resized to  $200 \times 200$  pixels

**Classes:**

1. Lamellar
2. Duplex
3. Acicular

**Data split:**

- Training: 1000 images
- Testing: 225 images

## Data Preprocessing

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**For Classification:**

- Image resizing
- Gaussian Blur ( $5 \times 5$  kernel)

**For Segmentation:**

- Median filtering
- CLAHE (contrast enhancement)

**Purpose:**

- \* Reduce noise
- \* Improve feature clarity

**Pipeline:**

**Stage 1: Classification**

- CNN model classifies image into 3 categories

**Stage 2: Segmentation**

- Apply specific algorithm based on class

## Classification

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**CNN Architecture**

- 3 Convolution layers
- 2 Max Pooling layers
- 1 Fully Connected layer

**Key features:**

- Activation: ReLU
- Optimizer: Adam
- Loss: Softmax cross-entropy

**Convolution Operation:**

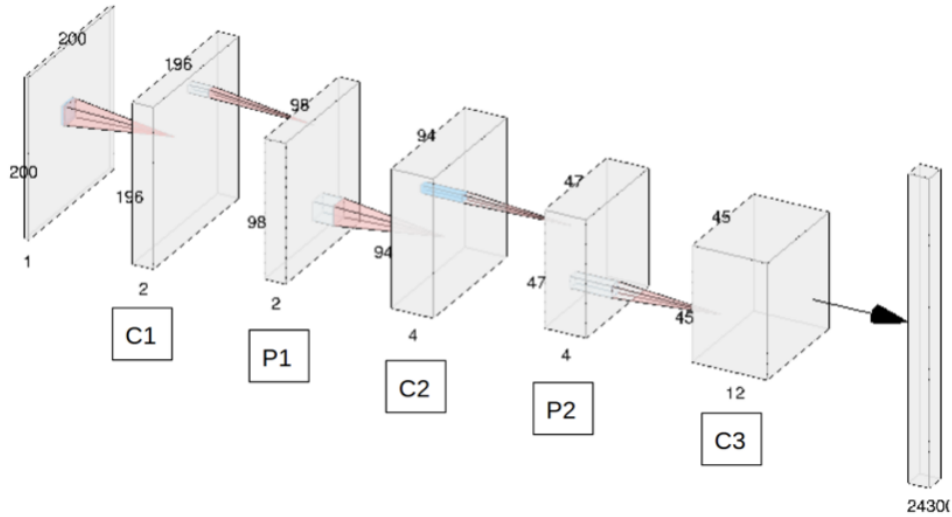
$$C(i, j) = \sum_m \sum_n I(i + m, j + n) F(m, n)$$

**ReLU Activation:**

$$f(x) = \max(0, x)$$

**Softmax Function:**

$$p_i = \frac{e^{a_i}}{\sum_k e^{a_k}}$$

**CNN Architecture**

The neural network employed is a convolutional neural network designed with three convolutional layers followed by two max-pooling layers and a single fully connected layer, where the input images are resized to 200×200 pixels and progressively downsampled through the network. Each convolutional layer uses small kernel filters (5×5 in the first two layers and 3×3 in the third) to extract spatial features at multiple scales, and the activation function used throughout the network is the Rectified Linear Unit (ReLU), defined as  $f(x) = \max(0, x)$ , which introduces non-linearity and helps avoid vanishing gradient issues. The final fully connected layer outputs class scores which are converted into probabilities using the softmax activation function, enabling multi-class classification. During training, the network parameters are initialized using Xavier initialization and optimized using the Adam optimizer with a learning rate of 0.001, while the loss is computed using softmax cross-entropy. Regularization is incorporated using L1 penalty

and early stopping based on validation accuracy to prevent overfitting. The model is trained on 1000 images with 200 used for validation over a maximum of 1500 iterations, and tested on 225 unseen images, achieving an average accuracy of approximately 94%, demonstrating effective learning and generalization.

Once the image is classified, it is taken into segmentation part where two main algorithms have been used:

### **1. Watershed Algorithm**

The watershed algorithm is used as a segmentation technique specifically for duplex microstructures to separate equiaxed grains from lamellar regions. It treats the image as a topographical surface where pixel intensities represent elevation, with low-intensity regions acting as valleys and high-intensity regions as peaks. Marker-based watershed is applied by first identifying markers corresponding to regions of interest, typically the grain interiors, and then flooding the image from these markers. As the flooding progresses, boundaries are formed where different regions meet, effectively segmenting the equiaxed grains. This allows the extraction of quantitative features such as the area fraction of equiaxed regions, which is important for understanding the microstructure.

**Used for:**

- Duplex microstructures

**Extracts:**

- Equiaxed grain area

### **2. Histogram of Oriented Gradients (HOG)**

The Histogram of Oriented Gradients (HOG) method is used for lamellar microstructures to capture texture and orientation information. Initially, gradient components in the x and y directions are computed using operators like Sobel filters, and from these, the gradient magnitude and orientation at each pixel are determined. A histogram is then constructed by grouping gradient orientations into bins, where peaks in the histogram correspond to dominant orientations in the image. In lamellar structures, strong directional patterns produce sharp peaks, which are used to identify and segment the dominant variant. By isolating pixels corresponding to these dominant orientations, a binary mask is created and overlaid on the original image to quantify features like the area fraction of the dominant phase. This method effectively captures the directional nature of lamellar microstructures and enables meaningful quantitative analysis.

**Used for:**

- Lamellar microstructures

→ **Extracts:**

- Dominant orientation
- Texture information

**Gradient Magnitude:**

$$G = \sqrt{G_x^2 + G_y^2}$$

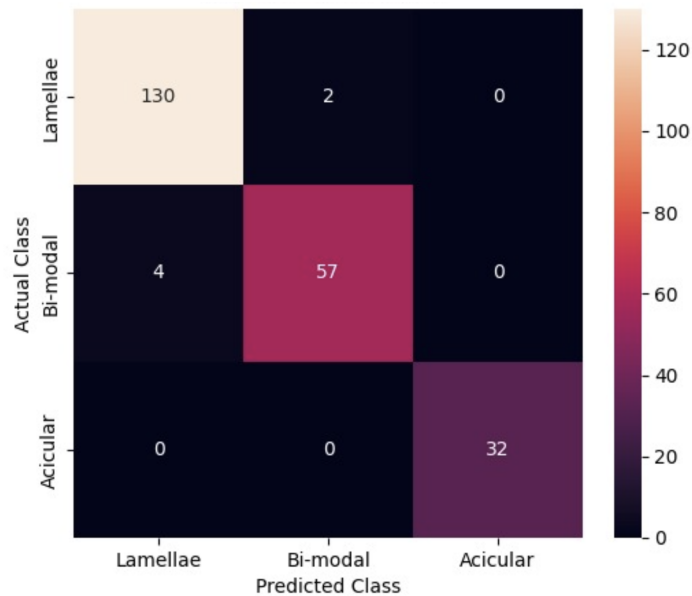
## Results

### Results based on classification

|                        |           |        |          |         |
|------------------------|-----------|--------|----------|---------|
| Classification Report: |           |        |          |         |
|                        | precision | recall | f1-score | support |
| Lamellae               | 0.97      | 0.98   | 0.98     | 132     |
| Bi-modal               | 0.97      | 0.93   | 0.95     | 61      |
| Acicular               | 1.00      | 1.00   | 1.00     | 32      |
| accuracy               |           |        | 0.97     | 225     |
| macro avg              | 0.98      | 0.97   | 0.98     | 225     |
| weighted avg           | 0.97      | 0.97   | 0.97     | 225     |

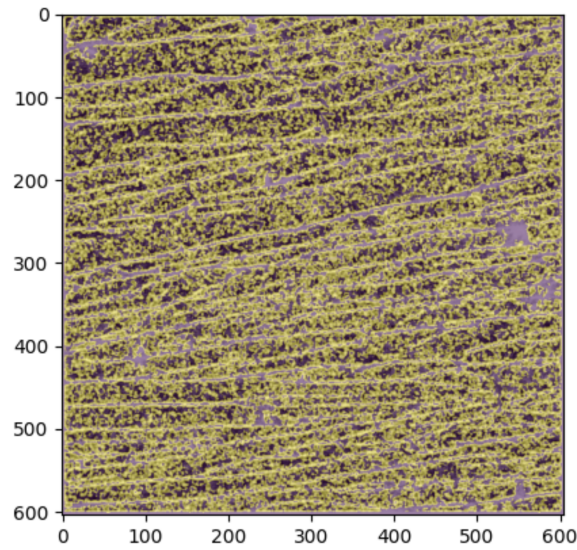
Best accuracy that we have acquired was 97%

**Confusion Matrix:** A table used to evaluate a classification model by comparing actual values with predicted values.

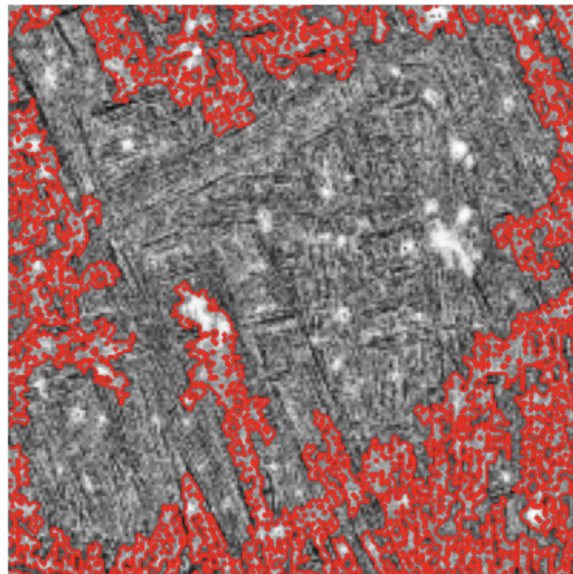


3x3 Confusion Matrix

## Results based on segmentation



Segmented lamella image



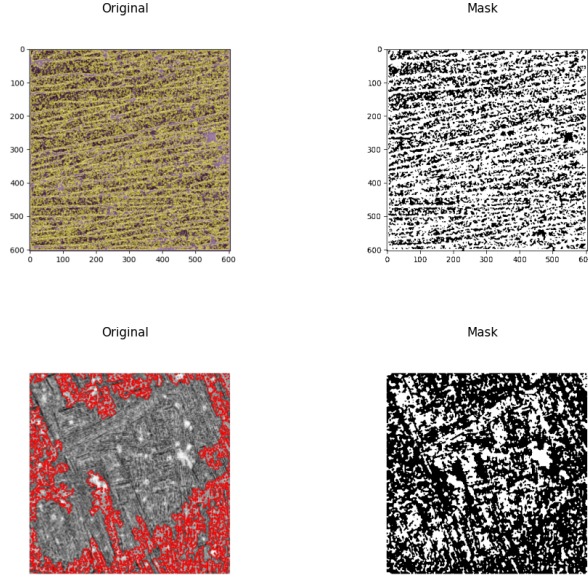
Segmented duplex image

## Quantification

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After segmentation, the watershed algorithm produces a labeled image where different integer values represent different regions. This output is then converted into a binary mask by selecting the region of interest, where pixels belonging to the desired phase are assigned a value of 255 and all other pixels are set to 0. This binary mask acts as a region-of-interest representation, isolating only the segmented phase from the background. The mask is stored as a NumPy array, allowing efficient pixel-wise operations. For quantification, the code performs a logical comparison to identify all foreground pixels (`mask == 255`),

effectively creating a boolean map of the phase. The number of these foreground pixels is computed and normalized by the total number of pixels in the image domain, giving an area-based metric. This process leverages digital image processing concepts such as binary segmentation, pixel intensity thresholding, and spatial domain analysis to convert the segmented output into a quantitative phase fraction.



Segmented Results

$$\text{Phase Fraction} = \frac{\text{White pixels}}{\text{Total pixels}} \times 100$$

*Result for Quantification*

```
Enter image path: seg_lamella.png

===== RESULT =====
Image: seg_lamella.png
Detected Class: Lamellae
Phase Fraction: 84.78%
```

## Conclusion

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In this project, a two-stage machine learning framework was successfully implemented for automated microstructure characterization of Ti-6Al-4V alloy. The convolutional neural network was able to accurately classify microstructures into lamellar, duplex, and acicular categories with high accuracy, demonstrating strong generalization capability on unseen data. Following classification, appropriate segmentation techniques such as watershed and Histogram of Oriented Gradients were applied to extract meaningful structural information. The segmented output was further used for quantification, where the phase fraction was computed based on pixel distribution. This approach significantly reduces

manual effort and provides a consistent and scalable solution for material characterization. Overall, the integration of deep learning with image processing techniques proves to be effective for analyzing complex microstructural patterns.

## Advantages and Disadvantages

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### Advantages

- Reduces manual inspection effort and human error in microstructure analysis, making the process faster and more consistent.
- Provides high accuracy in classification, improving reliability in quality control and material validation.
- Enables quantitative analysis such as phase fraction estimation, which is critical for predicting material properties.
- Scalable for large datasets, making it suitable for industrial environments with high data generation.
- Adaptable framework where models can be retrained for different alloys or microstructures.

### Disadvantages

- Requires a labeled dataset, and generating high-quality labeled microstructure images is time-consuming and expensive.
- Performance depends on image quality; variations in SEM conditions can affect accuracy.
- Segmentation methods are not fully generalizable and may require tuning for different materials.
- Requires computational resources for training and deployment, which may be a limitation for smaller setups.
- Limited interpretability, making it harder for engineers to fully trust decisions without validation.

## References

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1. Materials Today Communications, Elsevier. Available at: <https://www.sciencedirect.com/journal/materials-today-communications>
2. Advanced Steel Microstructural Classification by Deep Learning Methods, Scientific Reports, Nature.
3. Adaptive characterization of microstructure dataset using a two stage machine learning approach, Materials Today Communications, Elsevier. Available at: <https://www.sciencedirect.com/science/article/pii/S0927025620300847>